

Principles of Microeconomics

Week 3 · Day 2 — Repeated Games

Mehman Ismayilli

Yesterday cooperation collapsed in one shot. When rivals meet again and again, the *future* can rescue it.

Yesterday's puzzle

In the one-shot price war, cutting price is a dominant strategy, so both firms are stuck at (Low, Low) = (6, 6), giving up the better (High, High) = (10, 10).

	High	Low
High	(10, 10)	(2, 14)
Low	(14, 2)	(6, 6)

But firms meet again and again

A shop sets prices *every week*; airlines compete *every day*. If I undercut you today, you can punish me tomorrow.

The question

Can the **fear of future retaliation** sustain the good outcome (10, 10)? It depends on whether the game ends at a *known date* or could go on forever.

Learning goals — Day 2

By the end you can

- ① explain what a **repeated game** is, and how strategies can depend on *history*;
- ② show that a **finitely** repeated Prisoner's Dilemma *unravels* to defection;
- ③ value a stream of future payoffs using a **discount factor** δ ;
- ④ use a **grim-trigger** strategy to sustain cooperation when the game is repeated **forever**;
- ⑤ find the **critical discount factor** and explain what makes cooperation easier or harder.

What changes when a game is repeated

A repeated game

- The same **stage game** (e.g. the price war) is played in periods $t = 1, 2, 3, \dots$
- Players *see* what happened before, so a strategy can depend on **history**: “cooperate — *unless* you ever cheated me.”
- Payoffs add up across periods, with later ones discounted.

The key new possibility

Today's choice can *reward* or *punish* yesterday's behaviour. That is what makes cooperation *possible* — if the future matters enough.

Two very different worlds

A **known last period** versus a game that *might always continue*. They give opposite answers.

A known ending: the game unravels

Suppose the price war is played exactly $T = 10$ times, and both firms know it. **Start from the last period**

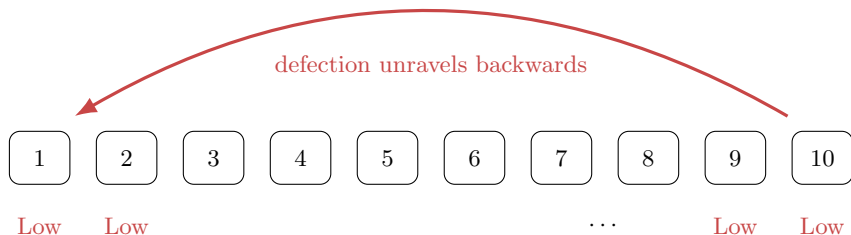
Period 10 is the *final* game — there is no future, so no way to punish. It is just the one-shot Prisoner's Dilemma: both play **Low**. **Now back up to period 9**

Both already know period 10 will be (Low, Low) *whatever* they do now. So period 9 has no future consequences either $\Rightarrow$ both play **Low**.

...and so on, back to period 1

The reasoning cascades backwards through every period. **Cooperation never happens** — both play Low every single round.

A known ending — the picture



The known last period is certain to be non-cooperative — and that certainty pins down every earlier period too.

The lesson of a known ending

Result

If the one-shot game has a single Nash equilibrium, then repeating it a *known, finite* number of times gives that same outcome in **every** period. A fixed deadline kills cooperation.

So what rescues cooperation?

Remove the *known* ending. If there is always a chance the game continues, backward induction has no place to start — and the future can be used to keep players honest.

Valuing the future: the discount factor

When the game may go on forever, we value future payoffs with a **discount factor** δ between 0 and 1:

$$\text{value} = \pi_1 + \delta \pi_2 + \delta^2 \pi_3 + \dots$$

Two ways to read δ

- *Patience*: \$1 next period is worth δ today. A patient player has δ close to 1.
- *Continuation*: δ can be the *probability the game continues* to the next round.

One handy formula

Getting the *same* amount π every period forever is worth

$$\pi + \delta\pi + \delta^2\pi + \dots = \frac{\pi}{1 - \delta}.$$

The grim-trigger strategy

Grim trigger

“Play **High** (cooperate) as long as everyone always has. The *first* time anyone plays Low, switch to **Low** forever.”

One cheat triggers a *permanent* price war — back to (6, 6) for the rest of time. A firm weighs two options:

Option 1 — keep cooperating

Earn 10 every period: $V_{\text{coop}} = \frac{10}{1 - \delta}$.

Option 2 — cheat today

Grab 14 now, then be punished with 6 forever after: $V_{\text{cheat}} = 14 + \frac{\delta \cdot 6}{1 - \delta}$.

When is cooperation worth it?

Cooperation holds when it is *not worth* cheating: $V_{\text{coop}} \geq V_{\text{cheat}}$,

$$\frac{10}{1-\delta} \geq 14 + \frac{6\delta}{1-\delta}.$$

Multiply through by $(1-\delta)$

$$10 \geq 14(1-\delta) + 6\delta = 14 - 8\delta.$$

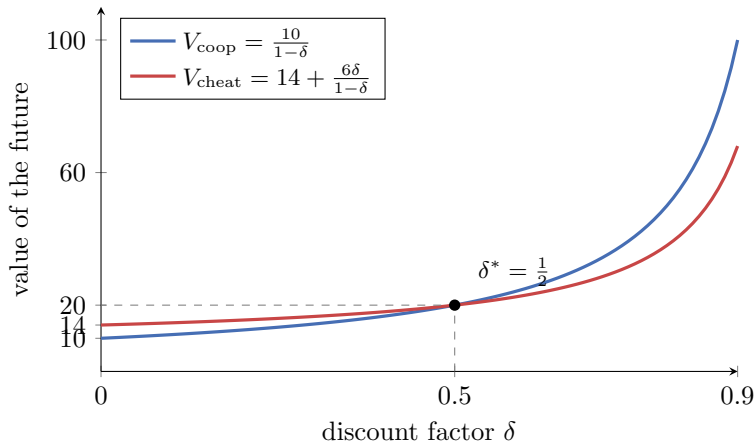
Solve for δ

$$8\delta \geq 4 \implies \boxed{\delta \geq \frac{1}{2}}.$$

If firms value the future enough ($\delta \geq \frac{1}{2}$), the threat of a permanent price war keeps them both at $(10, 10)$.

Numeric check: at $\delta = \frac{1}{2}$: $V_{\text{coop}} = \frac{10}{0.5} = 20$ and $V_{\text{cheat}} = 14 + \frac{6(0.5)}{0.5} = 20$ — *exactly a tie.* ✓

When is cooperation worth it? — the picture



For $\delta > \frac{1}{2}$ the cooperation line lies above — cheating is not worth it, and (10, 10) survives.

Reading the threshold

- $\delta \geq \frac{1}{2}$: patient firms, or a game very likely to continue $\Rightarrow$ cooperation holds.
- $\delta < \frac{1}{2}$: impatient firms, or a game likely to end soon $\Rightarrow$ the temptation to grab 14 today wins.

What makes cooperation *harder*?

- a bigger one-time prize from cheating;
- a weaker punishment (cheaters are not hurt much afterwards);
- interactions that are rare, or a deadline that is close.

Where you see this in the real world

Cooperation sustained by the future

- **Cartels** (e.g. OPEC): members hold output down, fearing a price war if anyone over-produces.
- **Firms and suppliers** keep promises to protect a long-term relationship.
- **Everyday trust**: neighbours, friends, repeat customers.

Why cartels still break down

When demand is about to fall, a member retires, or the “end” feels near, δ drops — and the temptation to cheat wins. This is also why competition authorities watch *stable, concentrated* markets most closely.

Known ending vs. no known ending — summary

	Known finite ending	No known ending
Can cooperation survive?	No — it unravels	Yes, if $\delta \geq \delta^*$
Why	the last period is certain	the future can punish cheating
How we solve it	backward induction	compare the value streams
Our example	Low in every period	High if $\delta \geq \frac{1}{2}$

The knife-edge is the *certainty of an ending*, not the number of rounds: it is a *known* last game that destroys cooperation, and its absence that restores it.

Day 2 checkpoint

You can now

- explain why a game with a *known* ending unravels to defection;
- value an endless stream of payoffs with $\frac{\pi}{1-\delta}$;
- test cooperation with a grim trigger and find the critical δ^* ;
- say what makes cooperation easier (patience, strong punishment) or harder.

Tomorrow — a different kind of failure

So far, trouble came from *strategic rivalry*. Day 3: markets that fail even with many players — **externalities, public goods, and the tragedy of the commons.**